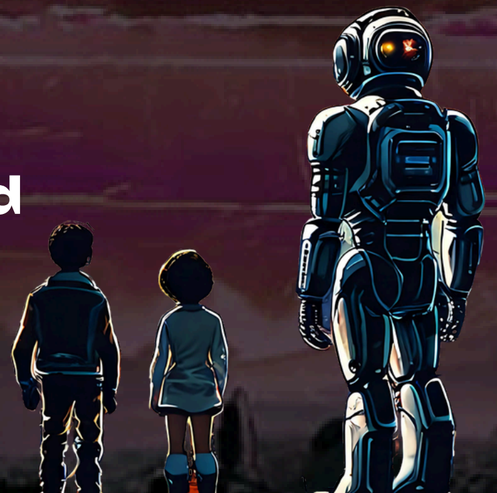
INDIA INNOVATES 2026

Team Name: **Code Blizzards**

Domain: **Digital Democracy**

VaaniAI - Multilingual AI Inbound & Outbound Calling Agent



Members:

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PROBLEM STATEMENT

100K+	Manual	No	22+
daily calls to MCD alone	grievance tracking today	24/7 availability today	Indian languages ignored

ROOT CAUSES:

- **Overloaded human agents** - cannot scale for 20 million Delhi residents
- **IVR systems** (press 1, press 2) are rigid, non-conversational, and frustrating
- **Language barrier** - Hindi, Punjabi, Urdu, Bhojpuri, English all spoken in Delhi
- **No memory** - citizens repeat their complaint on every call with no continuity
- **Zero real-time analytics** - no visibility into most common issues by ward
- **No proactive outreach** - surveys and reminders require massive manual effort

SOLUTION

An end-to-end AI-powered telephony platform for citizen services, grievance redressal, and outreach campaigns — enabling any government to deploy **intelligent, multilingual voice agents with contextual memory, real-time NLU, escalation handling, and analytics**.

Real-Time ASR

Indian-accent speech recognition in 10+ languages via Sarvam AI + Bhashini

Contextual NLU

LLM-powered intent understanding, entity extraction, and dialogue management

Citizen Memory

Vector DB stores conversation history — zero repeat context on callbacks

Smart Escalation

Emotion detection triggers seamless human handoff with full context transfer

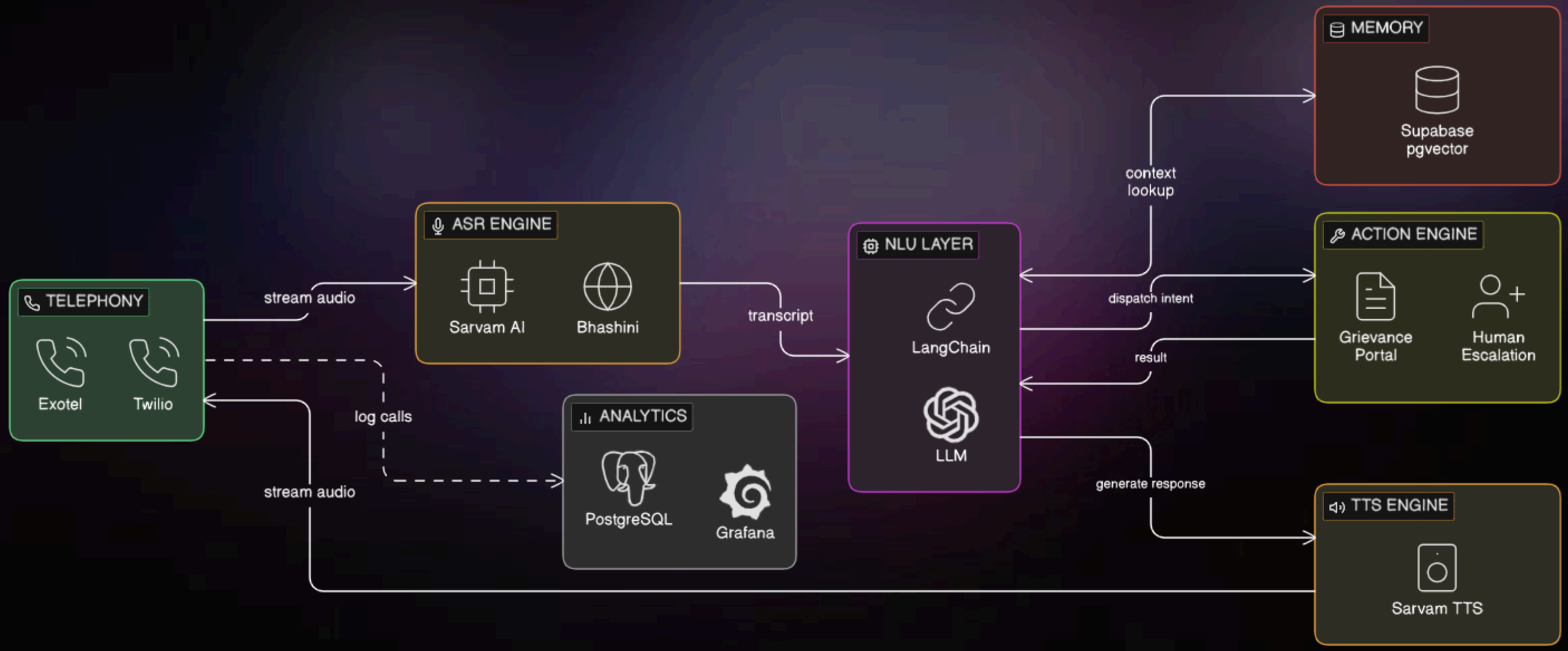
Live Analytics

Ward-level dashboards, sentiment trends, and anomaly detection in real time

Outbound Campaigns

10K+ simultaneous AI-driven outreach calls for surveys and reminders

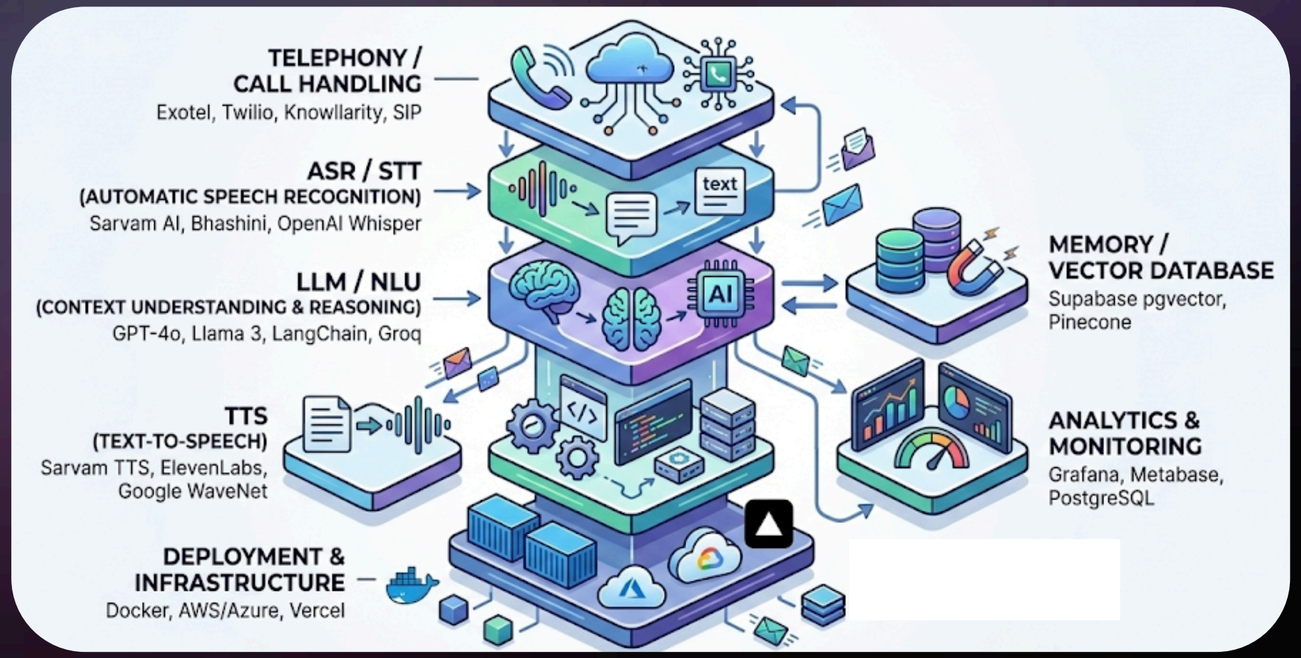
ARCHITECTURE



TECHNOLOGY USED

- **Telephony :** Exotel, Twilio, Knowlarity, SIP
- **ASR / STT :** Sarvam AI, Bhashini, OpenAI Whisper
- **LLM / NLU :** GPT-4o, Llama 3, LangChain, Groq
- **Memory :** Supabase pgvector, Pinecone
- **Channels:** SMS Gateway, Missed Call Service, Whatsapp

- **Text To Speech :** Sarvam TTS, ElevenLabs
- **Backend :** FastAPI, Python, WebSockets
- **Analytics :** Grafana, Metabase, PostgreSQL
- **Deployment :** Docker, AWS/Azure, Vercel



FEATURE / USP

- **India-First Multilingual AI:** Supports **12+ Indian languages**, including Hinglish, and integrates with Bhashini without custom training.
- **Real-Time Contextual Memory:** Stores citizen interactions for instant recall during callbacks.
- **Smart Escalation with Context Transfer:** Detects distress via **tone analysis** and transfers calls to human agents with conversation summaries.
- **Unlimited Scale Outbound:** Can make 10,000+ simultaneous calls for surveys and campaigns while gathering structured data.
- **Sub-2-Second Response Latency:** Fast conversations through streaming ASR and low-latency inference.
- **Live Ward-Level Analytics:** Real-time dashboards monitor complaint trends and sentiment scores for proactive responses.
- **Cost Efficiency:** Costs about ₹0.40 per call versus ₹18,000–₹25,000/month per human agent, achieving ROI in under 3 months at MCD scale.
- **Secure & Compliant Infrastructure:** Phone number-based caller verification, encrypted citizen data storage, role-based dashboard access, and full audit logs for every grievance action — built for government-grade trust.

EXTENDED FEATURES & IMPACT

- **WhatsApp Photo Integration:** Citizen receives a WhatsApp link post-call to attach photo/video evidence directly to their ticket
- **Missed Call Status Tracker:** Citizen gives a missed call to get an automatic callback with their ticket status — works on any phone, no app needed
- **Proactive Disaster Alerts:** Outbound calls to citizens in flood-prone wards during monsoon with ward-specific advisories
- **Post-Resolution Satisfaction Survey:** Automated callback after ticket closure to rate service quality — creates ward-level accountability scores
- **Duplicate Complaint Detection:** AI groups multiple complaints about the same issue into one high-priority ticket automatically
- **Vernacular FAQ Bot:** Citizens can ask general questions about services in any language without creating a ticket

Metric	Current State	With VaaniAI
Average Wait Time	Long wait times	< 5 seconds
Languages Supported	2–3 (Hindi, English)	12+ languages
24/7 Availability	No (8am–8pm only)	Yes, always-on
Grievance Acknowledgement	15–45 days	< 60 seconds
Monthly Call Capacity	~50,000 (staff-limited)	Unlimited
Outbound Campaigns	Manual, weeks to execute	10K calls in < 1 hour
Analytics	None / retrospective reports	Real-time, ward-level

REFERENCES / LINKS

Government / Official

- Bhashini: https://bhashini.gov.in
- MCD Grievance Portal: https://mcdonline.nic.in

Core AI Technologies

- Sarvam AI: https://sarvam.ai
- AI4Bharat: https://ai4bharat.org
- LangChain: https://langchain.com

Telephony

- Exotel: https://exotel.com
- Twilio: https://www.twilio.com/en-us

Research Papers

- IndicSUPERB benchmark: https://arxiv.org/abs/2208.11761
- IndicVoices : https://arxiv.org/abs/2403.01926

THANK YOU